

Richard A. Matos Arderí

COMPUTER SCIENTIST · DATA SCIENCE, ML & AI · FULL-STACK DEVELOPER

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TECHNICAL SKILLS

Languages

Python · Rust · TypeScript / JavaScript · R · C · SQL · LaTeX

Data Science & ML

scikit-learn · PyTorch · TensorFlow · XGBoost · LightGBM · CatBoost · SHAP · MLflow · NumPy · Pandas · SciPy

AI / NLP

LangChain · LangGraph · RAG · Multi-agent systems · AutoML · HuggingFace · Ollama · ChromaDB

Web & Full-stack

React · Next.js · Node.js · FastAPI · Tailwind · REST APIs · WebSockets

Mobile & Desktop

Android · iOS · cross-platform mobile (React Native) · desktop apps (Electron / Tauri)

Big Data & Distributed

Hadoop · HDFS · PySpark · Celery · Redis

DevOps & Tools

Docker · Git · GitHub Actions · PostgreSQL · MongoDB · Linux · pytest

LANGUAGES

Spanish Native

English B2 · Upper-intermediate

French A2 · Elementary

PROFILE

Computer Scientist trained at MATCOM (University of Havana), specializing in Data Science, Machine Learning and Artificial Intelligence, with strong full-stack and web development experience. Awarded the Scientific Merit Prize for an outstanding research record. Currently a Consultant Specialist at the Finlay Vaccine Institute (BioCubaFarma), where I design AI-powered platforms — multi-agent systems, RAG pipelines and AutoML workflows — and ship web, mobile and desktop applications. I take products end to end, from data and mathematical modeling through to polished, deployable software across web, Android, iOS and desktop.

PROFESSIONAL EXPERIENCE

Consultant Specialist

Mar 2025 – Present

Finlay Vaccine Institute (IFV) · BioCubaFarma

Data-science and AI consulting for vaccine development: epidemiological modeling, clinical-data integration and auditing, and the design of predictive models and supporting software within IFV and BioCubaFarma projects.

EDUCATION

B.Sc. in Computer Science

Jan 2023 – Jun 2026

Faculty of Mathematics and Computer Science (MATCOM), University of Havana

GPA: 4.71 / 5.00 · Cohort C411.

Diploma in Clinical Trials

2025

Center for Genetic Engineering and Biotechnology (CIGB) · BioCubaFarma

Specialized postgraduate training in the design, conduct, regulation and data management of clinical trials.

International Winter School

2026

University of Havana

Intensive program of international scientific training and exchange.

HONORS & AWARDS

- Scientific Merit Prize
Cuban Higher Education · 2026
- 1st Place · Honors Examination in Ordinary Differential Equations
MATCOM, UH · 2026
- “Relevant” paper at BIOQUIM 2025
Hybrid modeling of leptospirosis with vaccination

RESEARCH INTERESTS

- Biomedical AI
- Epidemiological Modeling
- Vaccine Development
- Mathematical Biology
- Distributed Systems

FEATURED PROJECTS

FinlAI AI · AutoML · Health
AI platform for optimizing vaccine development through integrated data management at the Finlay Vaccine Institute: multi-agent systems, RAG pipelines and AutoML workflows over clinical-trial data.

Hybrid Modeling ODE · ML · Epidemiology
System for epidemiological modeling combining differential-equation models with machine learning; applied to leptospirosis transmission and the evaluation of vaccination strategies.

AuditSoft Audit · Clinical data
Software for auditing clinical-trial data, ensuring quality, traceability and integrity of information within the IFV regulatory framework. Presented at COMPUMAT 2025.

ML Studio AutoML · Predictive models
Machine-learning modeling environment for predicting in-hospital mortality from AMI and performing inverse optimization of clinical variables, with interpretability (SHAP) and validation.

Tour Guide Cuba AI · NLP · Full-stack
Intelligent tourist-guide platform for Cuba with AI-powered recommendations and itinerary planning; team member on the full-stack application and its conversational assistant.

Logistic App Full-stack · Optimization
Logistics management application for route planning, order/fleet tracking and delivery optimization; team member working on core logic and the full-stack interface.

climaXtreme PySpark · HDFS · Docker
Analysis and modeling of extreme climate events over 8.6M+ records with Hadoop and PySpark; Streamlit dashboard, ML predictions and full Docker deployment.

CONFERENCE CONTRIBUTIONS

- “Hybrid Modeling: A system for epidemiological modeling”
XLI Scientific Conference, Faculty of Physics, UH · May 2026
- “AuditSoft: Software for auditing clinical-trial data”
COMPUMAT 2025 · International
- “ODE Mathematical Model for leptospirosis transmission”
COMPUMAT 2025 · International
- “Hybrid Modeling of leptospirosis transmission with vaccination strategies”
BIOQUIM 2025 · National · “Relevant” distinction
- “Interactive Web Application for predicting Acute Myocardial Infarction”
BIOQUIM 2025 · National · with A. Ponce González
- “ML models for in-hospital AMI mortality prediction and inverse optimization of clinical variables”
BIOQUIM 2025 · National · with A. Ponce González, A. Romero Imbert

RESEARCH PROJECTS

Optimizing vaccine development through integrated data management and AI

Finlay Vaccine Institute (IFV) · BioCubaFarma

Data Ecosystem Evolution Project

DICEI · IFV · BioCubaFarma

Development of a novel vaccine against Leptospirosis

IFV · BioCubaFarma

AMI registry with variable homogenization and quality indicators in Cuba

“Dr. Ernesto Guevara de la Serna” Teaching Hospital · MINSAP

VLIR-UOS: Integrated Real-Time Dengue Surveillance in Cuba for Outbreak Prediction

VLIR-UOS international cooperation